

PotholeWatch: An Open Data Commons for Road-Surface Health

A Concept Paper

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Executive Summary

Roads deteriorate faster than anyone can inspect them. In England and Wales a pothole has been filled every 18 seconds for a decade, the carriageway repair backlog stands near £17 billion, and pothole damage costs drivers more than £1 billion a year (ALARM, 2025). The vehicles and phones travelling those roads already sense every defect they cross—but each party that collects the data keeps it. Detection is a solved research problem; sharing is not.

This paper proposes PotholeWatch: an open data commons for road-surface health. Any device—a connected vehicle, a bus, a phone, a citizen with a web form—can contribute a tiny anonymous report (a timestamp, a location, a source class, an optional bump intensity). Reports cluster into defect signals published under an open licence that any council, researcher, or routing app can use. The system handles no repairs, no liability claims, and no money. It is infrastructure in the sense that OpenStreetMap is infrastructure: a neutral pool that exists because every private alternative fragments the data.

Each layer of this idea already exists in some closed, static, or single-jurisdiction form—citizen reporting sites, corporate vehicle-sensing pilots, per-council survey services, research prototypes. What does not exist is the open, machine-ingestible, cross-jurisdiction commons that lets them pool. This paper surveys that prior work, states precisely what is missing, and describes the smallest honest test of the concept. It is a proposal for discussion, not a specification; the hard problems—trust, equity, cold start, funding—are stated as open questions, not solved ones.

1.0 The Problem

A road defect is sensed many times a day. The bus that crosses it jolts. The delivery van's telematics log the shock. Drivers' phones register the same spike in their accelerometers. None of that reaches the council. Instead, the council learns about the defect from a periodic manual inspection, or from the small fraction of residents who navigate a reporting form—and by then the defect has grown, and a claim for a damaged wheel may already be in the post.

The scale is not in dispute. The Asphalt Industry Alliance's ALARM survey—the standard reference for local road maintenance in England and Wales—reports a carriageway repair backlog near £17 billion, with 1.9 million potholes filled in the last surveyed year at a cost of £137 million (ALARM, 2025). Vehicle damage from potholes costs UK drivers over £1 billion annually. Every highway authority runs the same loop: deterioration outpaces inspection, inspection outpaces repair budgets, and the data that could prioritise the spend sits unread in other people's sensors.

The failure is structural, and it is a data failure before it is an engineering one. Sensing is cheap and everywhere; the literature has treated accelerometer-based defect detection as a solved feasibility question since 2008 (Section 2.2). What is missing is a place where the sensed data can go—one that is not owned by a single vendor, locked to a single fleet, or scoped to a single council's boundary.

2.0 Prior Art

Road-condition sensing is a crowded field with twenty years of history. An honest proposal has to say what already exists and what, precisely, it claims to add.

2.1 Citizen Reporting

FixMyStreet, built by mySociety and running since 2007, lets residents report street problems on a map and routes each report to the responsible UK authority. It is open source and widely deployed. **Open311** standardises the interface between such reports and municipal issue-tracking systems, and is used by cities internationally. Both are human-centric by design: a person notices a problem, describes it, and submits it to one jurisdiction. Neither ingests machine-generated sensor streams, and neither produces a cross-jurisdiction open dataset of road condition.

2.2 Automatic Sensing Research

The Pothole Patrol project instrumented Boston taxis with accelerometers and GPS and showed that vehicle-mounted sensing could locate road anomalies with useful accuracy (Eriksson et al., 2008). Nericell demonstrated the same year that ordinary smartphones could detect potholes and braking events using their built-in sensors (Mohan et al., 2008). Mednis et al. (2011) achieved real-time detection on Android handsets with true-positive rates around 90% under favourable conditions. A steady stream of machine-learning refinements has followed. The feasibility question is closed: phones and vehicles can detect road defects. Every one of these projects, however, produced its own dataset for its own study area—none created an open, ongoing pool.

2.3 Street Bump and the Equity Lesson

Boston's Street Bump app (2011) crowdsourced pothole detection from residents' smartphones—and became a canonical case study in data bias. Smartphone ownership and app adoption skewed towards affluent neighbourhoods, so the potholes the system found skewed the same way, threatening to direct repair crews preferentially to wealthier streets (Crawford, 2013). Any crowdsourced road-condition system inherits this failure mode. It cannot be assumed away; it has to be designed for, and Section 4.3 treats it as an open design question.

2.4 Corporate Vehicle Data Pipelines

In April 2026, Waymo and Waze launched a pilot in which Waymo's autonomous vehicles detect potholes with their onboard sensors and share the locations with city governments through the Waze for Cities platform, submitting reports via Open311 in five US metropolitan areas (Waymo, 2026). This is machine-first road-defect sensing feeding municipalities at consumer scale—the closest live system to what this paper describes, and clear evidence that cities want the data. It is also structurally closed: one company's fleet, one company's platform, data shared with cities but not published as open data, and coverage bounded by where robotaxis operate. Several commercial

services occupy the adjacent per-council niche, surveying roads with smartphone-equipped municipal vehicles under paid contracts. In each case the data pipeline exists and the data stays private.

2.5 Open Geodata

OpenStreetMap holds surface and smoothness tags describing the general quality of road segments, but these are static attributes edited by mappers, not a stream of defect events, and they carry no freshness guarantees. Street-level imagery platforms and research datasets annotate surface quality from photographs. These projects prove that openly licensed road data sustains a contributor community; none of them tracks defects as they appear and disappear.

2.6 What Is Missing

Assemble the pieces and the gap is precise. Citizen reporting is human-only and per-jurisdiction. Research prototypes were study-bounded. Corporate pipelines are closed and fleet-bounded. Commercial surveys are paid and council-bounded. Open geodata is static. No system combines machine ingestion from any source, defect-level events with freshness, open licensing, and cross-jurisdiction scope. That combination—an open data commons for road-surface health—is what PotholeWatch proposes, and nothing found in the searches behind this paper currently provides it.

3.0 The Proposal

PotholeWatch is a pool, not a product. Its entire ambition is to be the neutral place road-surface observations go, so that any consumer—council, researcher, routing app, insurer—can draw on all of them instead of a private fraction.

3.1 A Minimal Report

The atomic unit is deliberately tiny [TARGET]: a unique identifier, a UTC timestamp, a WGS84 latitude and longitude, a source class (vehicle, IoT device, mobile app, or web form), an optional bump-intensity value from the reporting sensor, and an optional free-form metadata object. Roughly 120 bytes. No registration plates, no images required, no names, no accounts for basic reporting. Collecting little is the first line of privacy defence, but it is not sufficient on its own: a sequence of precise locations and timestamps from one device is a movement trail even without a name attached. Section 4.1 confronts that problem directly, because the trust mechanism makes it worse before design makes it better.

3.2 Machine-First Ingestion

The design assumption is that most reports arrive from machines: telematics units, dashcams with accelerometers, fleet management systems, phones running detection apps in the background. A human-facing web form exists and matters—it is how the system serves people whose streets have no instrumented traffic—but it is one client

among many, not the centre of the design. This inverts the citizen-reporting model of Section 2.1, and it is the load-bearing difference: machines report every defect they cross, on every trip, without motivation or form-filling.

3.3 Clustering and Freshness

Reports within roughly ten metres of one another merge into a defect cluster carrying a report count, a first-seen and last-seen time, and a source-diversity summary [TARGET]. Old reports age out, so the dataset tends to describe the road as it is rather than as it once was. Decay has a known blind spot: silence can mean a repaired road or merely an untravellered one, and a seasonal rural lane will look “fixed” every winter. Distinguishing repair from absence of traffic needs either traversal counts or council repair-log feedback, and which of those a deployment can obtain is a calibration question, as are the clustering thresholds themselves.

3.4 Open Data Out

Everything the commons aggregates is published under an open database licence of the kind OpenStreetMap uses (ODbL-style share-alike), with bulk export and a query interface. For councils, an Open311 export path means defect clusters can flow into existing municipal issue-tracking systems without new procurement. The commons does not tell a council what to fix; it hands over a prioritisation signal and steps back. There are no commissions, no exclusive access tiers, and no transaction role—the same neutrality argument that applies to DNS and to open mapping.

3.5 Why a Commons—and Why It Is Not OpenStreetMap

Every actor in Section 2 is currently rebuilding the same layer privately: Waymo for its fleet, each commercial vendor for its client councils, each research group for its study. The economics of a commons rest on ending that duplication—but the honest version of this argument has to concede two things first.

First, road-condition data is already a commercial asset. Vehicle manufacturers and map vendors sell hazard and surface data to one another today, so “nobody profits from hoarding it” is false at the OEM tier. The commons cannot expect donations from companies with a data business; its plausible contributors are the parties who own fleet data but have no data business—councils and their contractors. Local authorities already pay for telematics in their bus, refuse, and highways fleets. A data-sharing clause in those procurement contracts would route bump data to the commons at near-zero marginal cost, using leverage councils already hold [HYPOTHESIS].

Second, the OpenStreetMap precedent—the obvious analogy for any open geodata project—is weaker than it looks. OSM grew because mapping is a hobby people enjoy; nobody will hand-curate accelerometer spikes for pleasure. If the commons grows, it grows the way OSM’s bulk imports and corporate contributions did: institutions feeding it because policy or procurement tells them to, not volunteers feeding it for satisfaction. That is a narrower and more fragile adoption path, and it is why Section 6 proposes an institutional pilot rather than a public app launch.

4.0 Open Problems

Three hard problems stand between the concept and a working commons, and the first two pull against each other. The temptation is to present engineered solutions to them. They appear here as design questions with candidate approaches, because that is what they are.

4.1 Trust Versus Privacy—the Central Tension

An open ingestion endpoint invites junk, and in the UK the junk has a price tag. Pothole damage costs drivers over £1 billion a year in repairs, and a fraction of that returns to councils as compensation claims—small in aggregate, but paid per incident—so the single most predictable attack is fabricating a cluster of reports at a location shortly before filing a damage claim there. Other motives follow behind—triggering repairs on a favoured street, suppressing evidence of a hazard, or simple vandalism of the dataset.

The standard defence is device accountability: each device holds an attestation token, rate limits apply per token, one token gets one vote per cluster per day. But a persistent per-device token is a pseudonym, and a pseudonym attached to a stream of precise locations and timestamps is a movement trail—personal data under UK GDPR when the device rides in someone's pocket or private car. Taken naively, the trust mechanism destroys the privacy claim of Section 3.1. This tension is the central unsolved design problem of the commons, and this paper does not solve it.

Candidate mitigations exist and are stated here as hypotheses to test, not features [HYPOTHESIS]. Epoch-rotating anonymous credentials of the Privacy Pass family (RFC 9576) let a device prove “I am one real, rate-limited device” per short time window without any linkable long-term identifier. Publishing only aggregated clusters—never raw reports—and only once a diversity threshold is met would keep individual trails out of the public dataset, though not beyond inference everywhere: on a lane with a single regular traveller, even an aggregate cluster says who drove there. Aggregation reduces exposure; it does not abolish it. Fleet vehicles sidestep the issue, since a bin lorry's route is not private. Whether these measures survive both a competent adversary and a data-protection audit is exactly the kind of question the paper's Section 6 pilot exists to answer, and claims-context abuse is additionally blunted by licence terms: commons data supports triage and explicitly disclaims evidentiary use (Section 4.2). One corner of the threat model remains uncovered by design: the countless web form. Every mechanism above is device-based, so the form is the cheapest channel to abuse, and how form reports should be weighted, aggregated, or rate-limited is an open question—a deployment can sidestep it by running fleet-and-app-only, as the Section 6 pilot does.

4.2 Confidence, Not Truth

A cluster is a probabilistic signal that something is wrong with a road surface, never proof. Signals strengthen with source diversity—a cluster fed by a bus, a phone, and a van is harder to fake than one fed by three phones—and with reports spread over days

rather than seconds [HYPOTHESIS]. Intensity values need particular caution: an accelerometer reading is not comparable across a lorry, a saloon car, and a phone on a dashboard, so until per-source normalisation is demonstrated, intensity should be treated as presence evidence only, not severity. The proposed validation method is conventional: a deploying council physically inspects a sample of high-confidence and medium-confidence clusters, measures precision and recall, and tunes thresholds until the signal matches its inspection capacity. PotholeWatch data supports triage—where to send an inspector first. It does not support liability determinations, legal evidence, or safety-critical decisions, and the data licence should say so explicitly, which also removes most of the value of poisoning the dataset for a damage claim.

4.3 Equity

Street Bump's lesson (Section 2.3) applies in full: passive sensing follows device ownership, and device ownership follows wealth. The commons has one structural advantage the Boston app lacked—fleet vehicles—but the advantage must be claimed precisely. Buses run fixed arterial routes and never see most residential streets. Delivery vans go where parcels go, which correlates with spending power. The one fleet that genuinely visits every street is refuse collection: near-universal weekly coverage, on vehicles councils already operate, indifferent to neighbourhood income. A commons seeded with refuse-fleet data samples far more evenly than one built on volunteers' phones; whether that suffices, or whether deprivation-index reweighting is also needed, is an empirical question a pilot must answer. What the design must not do is publish a dataset whose known bias silently directs public repair budgets towards affluent streets.

5.0 Discussion and Limitations

The case against this concept is straightforward, and it deserves to be stated at full strength. The field's incumbents all hold distribution advantages the commons lacks: Waymo has a sensing fleet, Waze has drivers, FixMyStreet has council relationships, commercial vendors have contracts. A new commons starts with no fleet, no users, no funder, and a value proposition that only materialises after others contribute. Cold start killed better-resourced open projects than this one.

The counter-argument is coverage. No incumbent covers more than a sliver: five US metros for the Waymo pilot, individual client councils for the vendors, one jurisdiction per FixMyStreet instance. England alone has 153 local highway authorities (NAO, 2024), most of which will never appear in any corporate pilot. For them the choice is not "commons versus incumbent" but "commons versus nothing". Open projects tend to win precisely where proprietary coverage is uneconomic—with the caveat, argued in Section 3.5, that this commons cannot rely on OpenStreetMap's volunteer engine and must recruit institutions instead. The claim remains a hypothesis until a pilot produces evidence [HYPOTHESIS].

Sustainability is unsolved. Hosting a global ingestion endpoint and serving open data costs real money, the system by design earns none, and no institution has been approached. Candidate models—public-sector sponsorship from authorities that

consume the data, foundation hosting on the OpenStreetMap pattern, in-kind infrastructure from a university or civic-tech body—are listed here as possibilities, not plans.

Finally, nothing in this paper is built. Earlier versions of this document described connectors, training programmes, service levels, and audit regimes as though they existed; they did not, and do not. What exists is a data model sketch, a survey of a crowded field, and an argument that the missing piece is worth building. Readers should weigh the concept accordingly.

6.0 A First Test

The concept needs adversaries before it needs code. The contributors sought are geospatial and distributed-systems engineers who can attack the ingestion and clustering model, privacy engineers who can judge whether the Section 4.1 mitigations survive a data-protection audit, council asset-management practitioners who can say whether an open triage signal would actually change their repair scheduling, fleet operators who can say what it would take to share telematics bump data, and open-data governance people who can say whether ODbL-style licensing fits sensor-derived data.

If the concept survives that scrutiny, the smallest honest test is one borough, one refuse fleet: retrofit accelerometer loggers to a council's own bin lorries—vehicles that already visit every street weekly and carry no passenger-privacy burden—and publish open defect clusters for one collection area, ground-truthed against the council's inspections for one season. The pilot has defined success criteria [TARGET]: at least 70% precision on physically inspected high-confidence clusters over a sample of one hundred or more; weekly report coverage of at least 80% of the area's street-kilometres; and a stated go/no-go decision at season's end. Precision is gated while recall is only measured, deliberately: false alarms destroy a council's trust in the signal faster than missed defects, which today's inspection regimes miss anyway. Its principal cost drivers are the sensor retrofit, a modest hosting bill, and inspector time for the ground-truth sample; a full costing is deliberately left to the pilot's sponsoring authority, because inventing figures here would repeat the sin this paper was rewritten to remove.

Success would look unremarkable: a public map of probable defects, a council fixing the worst of them a little sooner, and a dataset anyone can build on. Roads are public. The data describing their condition should be too.

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Appendix A: Document Metadata

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Appendix B: Glossary

Term	Definition
ALARM	Annual Local Authority Road Maintenance survey, published by the Asphalt Industry Alliance
API	Application Programming Interface
GDPR	General Data Protection Regulation—EU/UK data protection law
GPS	Global Positioning System
IoT	Internet of Things—network-connected sensing devices
Lidar	Light detection and ranging—laser-based distance sensing
ODbL	Open Database License—the share-alike open data licence used by OpenStreetMap
Open311	A standardised interface for civic issue reporting to municipal systems
OSM	OpenStreetMap—the collaboratively edited open world map
Telematics	In-vehicle systems that record and transmit vehicle sensor data
UTC	Coordinated Universal Time
WGS84	World Geodetic System 1984—the coordinate reference system used by GPS

Appendix C: Methodology and Transparency Statement

Author Context and Expertise

This work is grounded in 30 years of multi-sector experience in Systems Engineering. The analytical framework reflects professional history, focusing on practical application.

Digital Methodology and Accessibility

In the spirit of transparency, I utilised a suite of Generative AI tools—specifically Claude, Google Gemini, and ChatGPT—to assist in the production of this paper. These tools were employed as “force multipliers” for data synthesis and editorial accessibility. As a writer with dyslexia, I leverage these models as assistive technology to refine grammar and streamline sentence structure.

Verification and Integrity

While AI assisted in processing data, the intellectual oversight is entirely human. I performed manual validation of all citations and accept full responsibility for the accuracy and originality of the final output.

Appendix D: References

Citation	Reference
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